

Signal processing layer

1 Notation and setup

A single trial is a matrix

$$X \in \mathbb{R}^{C \times T},$$

C channels by T time samples, already band-pass filtered and baseline-corrected. Rows are electrodes, columns are samples. Write X_i for the i th channel's time course.

A calibration set is $\{(X_n, y_n)\}_{n=1}^N$ with $y_n \in \{0, 1\}$. All quantities below that are *learned* are fitted on training folds only and then frozen; applying them to a test trial uses the stored values.

2 Covariance

2.1 Definition

Assume each row is mean-centred, $\sum_t X_{it} = 0$. The sample covariance is

$$\Sigma = \frac{1}{T-1} X X^\top \in \mathbb{R}^{C \times C}, \quad \Sigma_{ij} = \frac{1}{T-1} \sum_{t=1}^T X_{it} X_{jt}.$$

The diagonal holds per-channel variance; off-diagonals hold covariation between channel pairs.

2.2 Two properties

Symmetry. $\Sigma_{ij} = \Sigma_{ji}$, immediate from the definition, so $\Sigma = \Sigma^\top$.

Positive semi-definiteness. For any $w \in \mathbb{R}^C$,

$$w^\top \Sigma w = \frac{1}{T-1} w^\top X X^\top w = \frac{1}{T-1} \|X^\top w\|_2^2 \geq 0.$$

The middle expression is the variance of the mixed signal $w^\top X$, which cannot be negative. It is strictly positive, giving $\Sigma \in \mathcal{S}_{++}^C$, whenever no linear combination of channels is identically zero — requiring $T \geq C$ and no exactly redundant channel.

2.3 Shrinkage

For short windows the empirical estimator is poorly conditioned, which makes the matrix logarithm of Section 4 numerically unstable. Use the Ledoit–Wolf estimator

$$\hat{\Sigma} = (1 - \alpha) \Sigma + \alpha \frac{\text{tr}(\Sigma)}{C} I, \quad \alpha \in [0, 1]$$

with α chosen analytically. This pulls eigenvalues away from zero, and hence their logarithms away from $-\infty$.

2.4 Time invariance — why this is not yet enough

Let P be any $T \times T$ permutation. Then

$$(XP)(XP)^\top = XPP^\top X^\top = XX^\top,$$

so Σ is invariant to reordering the time samples. It encodes *spatial* structure only. A response defined by its shape and latency is invisible to it, which is what Section 3 repairs.

3 Spatial filtering and prototype augmentation

3.1 xDAWN

A spatial filter $w \in \mathbb{R}^C$ produces a virtual channel $w^\top X$. Let $\bar{X}_1 \in \mathbb{R}^{C \times T}$ be the average of the class-1 trials (the evoked response, estimated by least squares under a superposition model), and let Σ_{sig} and Σ_{tot} be the covariances of $\bar{X}_1$ and of the raw data. xDAWN maximises the Rayleigh quotient

$$w^\star = \arg \max_w \frac{w^\top \Sigma_{\text{sig}} w}{w^\top \Sigma_{\text{tot}} w},$$

whose stationary points solve the generalized eigenvalue problem $\Sigma_{\text{sig}} w = \lambda \Sigma_{\text{tot}} w$. Keeping the top K eigenvectors gives $W \in \mathbb{R}^{C \times K}$.

Two consequences worth stating. Weights may be negative, which lets the filter subtract channels carrying shared noise — something no single-electrode choice can do. And hard selection is in the feasible set ($w = e_i$), so a learned filter cannot do worse than hand-picking except through estimation error.

3.2 Augmentation

Stack the filtered prototype on the filtered trial:

$$\tilde{X} = \begin{bmatrix} W^\top \bar{X}_1 \\ W^\top X \end{bmatrix} \in \mathbb{R}^{2K \times T}, \quad \Sigma = \frac{1}{T-1} \tilde{X} \tilde{X}^\top \in \mathcal{S}_{++}^{2K}.$$

In block form,

$$\Sigma = \begin{bmatrix} \Sigma_{\bar{X}\bar{X}} & \Sigma_{\bar{X}X} \\ \Sigma_{X\bar{X}} & \Sigma_{XX} \end{bmatrix}.$$

The off-diagonal block has entries

$$(\Sigma_{\bar{X}X})_{ij} = \frac{1}{T-1} \sum_t (W^\top \bar{X}_1)_{it} (W^\top X)_{jt},$$

a correlation of the trial against a template with fixed time structure — a matched filter. Permuting the trial’s samples destroys it. The timing information lost in Section 2.4 re-enters here.

The diagonal block Σ_{XX} carries the trial’s own noise level, so the classifier can weigh template match against the noise it sat in.

Oscillatory variant. If the target is a sustained rhythm rather than a time-locked response, drop the augmentation: band-pass to the band of interest and take $\Sigma = \frac{1}{T-1} XX^\top$ directly. Everything from Section 4 onward is unchanged. Time invariance is then a feature, not a defect.

4 The manifold

4.1 Why not Euclidean

$\mathcal{S}_{++}^d$ is an open convex cone inside the vector space of symmetric matrices, not a subspace: it is not closed under negation or subtraction. Treating it as flat distorts results. With $A = \text{diag}(1, 4)$ and $B = \text{diag}(4, 1)$, both of determinant 4,

$$\frac{1}{2}(A + B) = \text{diag}(2.5, 2.5), \quad \det = 6.25,$$

spread that neither input had. Covariance composes multiplicatively: scaling a signal by a scales its covariance by a^2 . Arithmetic averaging is a rule for additive quantities.

4.2 Matrix functions

For $M \in \mathcal{S}_{++}^d$ with eigendecomposition $M = V\Lambda V^\top$, $\Lambda = \text{diag}(\lambda_1, \dots, \lambda_d)$, $\lambda_i > 0$, define

$$M^p = V\Lambda^p V^\top, \quad \log M = V(\log \Lambda)V^\top, \quad \exp S = V(\exp \Lambda)V^\top.$$

Eigenvectors are untouched; the scalar function acts on eigenvalues. Positive definiteness is exactly the condition making $\log M$ real and unique. Note this is *not* the entrywise logarithm.

4.3 Affine-invariant metric

$$\delta(A, B) = \|\log(B^{-1/2}AB^{-1/2})\|_F = \left(\sum_i \log^2 \lambda_i\right)^{1/2},$$

where λ_i solve $\det(A - \lambda B) = 0$. For any invertible $G \in \mathbb{R}^{d \times d}$,

$$\delta(GAG^\top, GBG^\top) = \delta(A, B).$$

Volume conduction and re-referencing are linear maps on the sensors, so this geometry is blind to an entire class of nuisance transformations.

4.4 Geometric mean

$$M = \arg \min_{M \in \mathcal{S}_{++}^d} \sum_{n=1}^N \delta^2(M, \Sigma_n).$$

No closed form for $d > 1$. Fixed-point iteration, from $M^{(0)} = \frac{1}{N} \sum_n \Sigma_n$:

$$S^{(k)} = \frac{1}{N} \sum_{n=1}^N \log(M^{(k)-1/2} \Sigma_n M^{(k)-1/2}), \quad M^{(k+1)} = M^{(k)1/2} \exp(S^{(k)}) M^{(k)1/2},$$

stopping when $\|S^{(k)}\|_F < \varepsilon$. The mean is characterised by its tangent-space mean being zero.

5 Tangent space

5.1 The map

Fix the reference M from Section 4.4. For a trial covariance Σ :

1. Whiten. $\tilde{\Sigma} = M^{-1/2} \Sigma M^{-1/2}$.

The two-sided form is forced: a linear map $x \mapsto Gx$ on the data sends $\Sigma \mapsto G\Sigma G^\top$, since covariance is a quadratic form. With $G = M^{-1/2}$ (symmetric), this is the sandwich. One-sided $M^{-1}\Sigma$ has the same eigenvalues but is not symmetric and leaves $\mathcal{S}_{++}^d$.

Check: $\Sigma = M \Rightarrow \tilde{\Sigma} = I$. The eigenvalues of $\tilde{\Sigma}$ are the generalized eigenvalues of (Σ, M) : the ratio of this trial's variance to typical, per direction, and so independent of overall gain.

2. Logarithm. $S = \log \tilde{\Sigma}$.

Now $\Sigma = M \Rightarrow S = 0$: the reference is the origin. A direction with $4\times$ typical variance maps to $+\log 4$, with $\frac{1}{4}\times$ to $-\log 4$. Multiplicative becomes additive.

3. Vectorize. With $\mathcal{I} = \{(i, j) : i \leq j\}$,

$$v = \text{vec}(S) = [S_{11}, \sqrt{2} S_{12}, \dots, S_{22}, \dots] \in \mathbb{R}^{d(d+1)/2}.$$

The $\sqrt{2}$ on off-diagonals is required for the isometry $\|v\|_2 = \|S\|_F$, since each off-diagonal entry appears twice in $\|S\|_F^2$. Hence $\|v\|_2 = \delta(\Sigma, M)$: Euclidean distance on the chart equals manifold distance from the reference.

For $K = 3$ filters, $d = 2K = 6$ and $\dim v = 21$, against $C \cdot T$ raw features — the dimensionality reduction that makes the model trainable at a few hundred trials.

5.2 Interpretation

The chart is exact at M and degrades with distance, like a flat map of a curved surface: accurate for a city, distorting for a hemisphere. Since M is the mean of the training data, most trials sit near the point of tangency.

6 Classifier

Standardize each coordinate using training-fold statistics only: $z = (v - \mu) \odot \sigma$. The chart is already centred near zero by construction, so this is about scale: diagonal entries are log-variance ratios, off-diagonals carry the $\sqrt{2}$ weight, and the ℓ_2 penalty is not scale invariant.

$$\hat{\beta} = \arg \min_{\beta, \beta_0} \sum_{n=1}^N c_{y_n} \log(1 + e^{-\tilde{y}_n(\beta^\top z_n + \beta_0)}) + \frac{1}{2C_{\text{reg}}} \|\beta\|_2^2,$$

with $\tilde{y}_n \in \{-1, +1\}$ and c_y inverse class frequencies.

ℓ_2 , **not** ℓ_1 . Covariance coordinates are strongly correlated; ℓ_1 retains an arbitrary member of a correlated group and zeroes the rest, giving coefficients unstable across folds. There is no sparsity to gain at 21 features.

C_{reg} **must be retuned at every training size**. The optimum depends on N . Fixing it while sweeping N measures a curve about a mis-set hyperparameter, not a learning curve. Use an inner cross-validation loop.

Score, not decision. The output $s = \sigma(\hat{\beta}^\top z + \hat{\beta}_0)$ is used to rank candidate windows. Deployment class imbalance ($\approx 1:60$) is far from calibration imbalance ($\approx 1:5$), so s is badly calibrated as a probability; ranking requires only monotonicity in the likelihood ratio, which is preserved.

7 Summary

Fit (once per subject per session).

1. Filter, epoch, reject artifacts.
2. Learn W and $\bar{X}_1$ from labeled trials.
3. Per trial: augment, estimate Σ_n with shrinkage.
4. Iterate for M ; store $M^{-1/2}$.
5. Map all trials to v_n ; fit μ, σ and $\hat{\beta}$, tuning C_{reg} by inner cross-validation.

Score. Same W , $\bar{X}_1$, $M^{-1/2}$, μ , σ , $\hat{\beta}$ — all frozen. Whiten, log, vectorize, standardize, predict, rank.

Leakage. Recomputing M (or W , $\bar{X}_1$, μ , σ) on test data both leaks labels and places train and test features in different coordinate systems, voiding the fitted weights.

Verification. The following were checked numerically against an independent implementation and all hold to $\sim 10^{-10}$: $M^{-1/2}MM^{-1/2} = I$; $\text{vec}(\log(M^{-1/2}MM^{-1/2})) = 0$; $\|v\|_2 = \delta(\Sigma, M)$; affine invariance under random invertible G ; and $\frac{1}{N} \sum_n v_n = 0$ at the geometric mean.